

DATE:25-07-2026

EXPLICIT CURSORS

1. Write a program that will prompt for a JOB and delete the employees working in that Job.

Store the number of records deleted in a session variable.

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> VARIABLE deleted_count NUMBER;
SQL>
SQL> DECLARE
  2   v_job EMP.JOBTITLE := '&JOB';
  3 BEGIN
  4   DELETE FROM EMP
  5   WHERE JOB = v_job;
  6   :deleted_count := SQL%ROWCOUNT;
  7   DBMS_OUTPUT.PUT_LINE(SQL%ROWCOUNT || ' employee(s) deleted. ');
  8
  9   COMMIT;
 10 END;
 11 /
Enter value for job: CLERK
old 2:   v_job EMP.JOBTITLE := '&JOB';
new 2:   v_job EMP.JOBTITLE := 'CLERK';
3 employee(s) deleted.

PL/SQL procedure successfully completed.
```

2. Rollback the previous Delete Statement.

```
SQL> ROLLBACK;

Rollback complete.
```

3. Write a program that displays the Top N salaries using simple Explicit Cursors.

```
SQL> DECLARE
  2   v_n NUMBER := 6N;
  3
  4   CURSOR emp_cursor IS
  5     SELECT EMPNO, ENAME, SAL
  6     FROM EMP
  7     ORDER BY SAL DESC;
  8
  9   v_empno EMP.EMPNO%TYPE;
 10   v_ename EMP.ENAME%TYPE;
 11   v_sal EMP.SAL%TYPE;
 12   v_count NUMBER := 0;
 13
 14 BEGIN
 15   OPEN emp_cursor;
 16
 17   LOOP
 18     FETCH emp_cursor INTO v_empno, v_ename, v_sal;
 19
 20     EXIT WHEN emp_cursor%NOTFOUND OR v_count = v_n;
 21
 22     DBMS_OUTPUT.PUT_LINE(
 23       'EMPNO: ' || v_empno ||
 24       ' ENAME: ' || v_ename ||
 25       ' SALARY: ' || v_sal
 26     );
 27     v_count := v_count + 1;
 28   END LOOP;
 29
 30   CLOSE emp_cursor;
 31
 32 END;
 33 /
Enter value for n: 10
old 2:   v_n NUMBER := 6N;
new 2:   v_n NUMBER := 10;
EMPNO: 7189 NAME: VAISH SALARY: 50000
EMPNO: 7102 NAME: FORD SALARY: 3000
EMPNO: 7118 NAME: SCOTT SALARY: 3000
EMPNO: 7566 NAME: JONES SALARY: 2975
```

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4.Create a Simple Explicit Cursor that display all the employees who are earning more than the average salary of Emp Table.

```

1  Command Prompt - sqlplus / * *
2
3  SQL> SET SERVEROUTPUT ON;
4  SQL>
5  SQL> DECLARE
6  2      v_avg_sal NUMBER;
7  3
8  4      CURSOR emp_cursor IS
9  5          SELECT EMPNO, ENAME, SAL
10 6              FROM EMP
11 7              WHERE SAL > v_avg_sal;
12 8
13 9      v_empno EMP.EMPNO%TYPE;
1410      v_ename EMP.ENAME%TYPE;
1511      v_sal    EMP.SAL%TYPE;
1612
1713 BEGIN
1814     -- Find average salary
1915     SELECT AVG(SAL)
2016     INTO v_avg_sal
2117     FROM EMP;
2218
2319     DBMS_OUTPUT.PUT_LINE('Average Salary: ' || v_avg_sal);
2420     DBMS_OUTPUT.PUT_LINE('Employees earning more than average salary:');
2521
2622     OPEN emp_cursor;
2723
2824     LOOP
2925         FETCH emp_cursor INTO v_empno, v_ename, v_sal;
3026
3127         EXIT WHEN emp_cursor%NOTFOUND;
3228
3329         DBMS_OUTPUT.PUT_LINE(
3430             'EMPNO: ' || v_empno ||
3531             ' NAME: ' || v_ename ||
3632             ' SALARY: ' || v_sal
3733         );
3834     END LOOP;
3935
4036     CLOSE emp_cursor;
4137
4238 END;

```

[illegible]

5. Create a Simple Explicit cursor that display the employees who earn more than 2000 and have joined the company after 15-jun-1981.

Print the output : $\diamond$ earns $\diamond$ and joined the organization on $\diamond$.

```

1  Command Prompt - sqlplus / X + ->
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
2
3     CURSOR emp_cursor IS
4         SELECT ENAME, SAL, HIREDATE
5             FROM EMP
6             WHERE SAL > 2000
7               AND HIREDATE > '15-JUN-1981';
8
9     v_ename EMP.ENAMETYPE;
10    v_sal    EMP.SALTYPE;
11    v_date   EMP.HIREDATETYPE;
12
13 BEGIN
14
15     OPEN emp_cursor;
16
17     LOOP
18         FETCH emp_cursor INTO v_ename, v_sal, v_date;
19
20         EXIT WHEN emp_cursor%NOTFOUND;
21
22         DBMS_OUTPUT.PUT_LINE(
23             v_ename || ' earns ' || v_sal ||
24             ' and joined the organization on ' || v_date
25         );
26
27     END LOOP;
28
29     CLOSE emp_cursor;
30
31 END;
32 /
SCOTT earns 3800 and joined the organization on 19-APR-87
FORD earns 3000 and joined the organization on 03-DEC-81
VAISHI earns 5000 and joined the organization on 19-JUL-26

PL/SQL procedure successfully completed.

```

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6.Create a Simple Explicit cursor that displays the hiredate in the format DD-MM-RRRR and Day. Sort the records on DAY starting from Saturday.

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
2
3     CURSOR emp_cursor IS
4         SELECT ENAME,
5                TO_CHAR(HIREDATE, 'DD-MM-RRRR') AS H_DATE,
6                TO_CHAR(HIREDATE, 'DAY') AS H_DAY
7         FROM EMP
8         ORDER BY MOD(TO_CHAR(HIREDATE, 'D') + 1, 7);
9
10    v_ename EMP.ENAME%TYPE;
11    v_date  VARCHAR2(20);
12    v_day   VARCHAR2(10);
13
14 BEGIN
15
16     OPEN emp_cursor;
17
18     LOOP
19         FETCH emp_cursor INTO v_ename, v_date, v_day;
20
21         EXIT WHEN emp_cursor%NOTFOUND;
22
23         DBMS_OUTPUT.PUT_LINE(
24             v_ename || ' joined on ' || v_date ||
25             ' ' || v_day || ')';
26     );
27
28 END LOOP;
29
30 CLOSE emp_cursor;
31
32 END;
33 /
ALLEN joined on 20-02-1981 (FRIDAY )
BLAKE joined on 01-05-1981 (FRIDAY )
VAISH joined on 19-07-2026 (SUNDAY )
SCOTT joined on 19-08-1987 (SUNDAY )
WARD joined on 22-02-1981 (SUNDAY )
```

7.Create a simple explicit cursor that display the employees names and commission amounts. If an employee does not earn commission, show 'NO Commission' .Label the COMM.

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
2
3     CURSOR emp_cur IS
4         SELECT ename,
5                NVL(TO_CHAR(comm), 'NO Commission') AS comm
6         FROM emp;
7
8     v_ename EMP.ename%TYPE;
9     v_comm  VARCHAR2(20);
10
11 BEGIN
12     OPEN emp_cur;
13
14     DBMS_OUTPUT.PUT_LINE(RPAD('EMPLOYEE NAME', 20) || 'COMM');
15
16     LOOP
17         FETCH emp_cur INTO v_ename, v_comm;
18         EXIT WHEN emp_cur%NOTFOUND;
19
20         DBMS_OUTPUT.PUT_LINE(RPAD(v_ename, 20) || v_comm);
21     END LOOP;
22
23 CLOSE emp_cur;
24
25 END;
26 /
EMPLOYEE NAME      COMM
ALLEN              380
WARD               500
JONES              NO Commission
MARTIN             1400
BLAKE              NO Commission
SCOTT              NO Commission
TURNER             0
FORD               NO Commission
VAISH              NO Commission

PL/SQL procedure successfully completed.

SQL>
```

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8.Create a parameter cursor that takes DEPTNO as a parameter and update the star column with '*'.
Every '*' represents a 1000. For example : if the employees salary is 5000 then update star with '*****'. Use Parameter Cursor, CURSOR WITH FOR LOOP and WHERE CURRENT OF OPTION.

```
SQL> ALTER TABLE EMP
2 ADD STAR VARCHAR2(20);

Table altered.

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
2  CURSOR emp_cur(p_deptno NUMBER) IS
3  SELECT sal, star
4  FROM emp
5  WHERE deptno = p_deptno
6  FOR UPDATE OF star;
7 BEGIN
8  FOR rec IN emp_cur(&DEPTNO)
9  LOOP
10 UPDATE emp
11 SET star = RPAD('*', TRUNC(rec.sal/1000), '*')
12 WHERE CURRENT OF emp_cur;
13 END LOOP;
14
15 COMMIT;
16 DBMS_OUTPUT.PUT_LINE('STAR column updated successfully.');
```

Enter value for deptno: 10
old 8: FOR rec IN emp_cur(&DEPTNO)
now 8: FOR rec IN emp_cur(10)
STAR column updated successfully.
PL/SQL procedure successfully completed.

9.Write a Parameter Cursor program that promotes CLERK who earn more than 1000 to SR CLERK and increase the salary by 10%. Pass CLERK as a parameter to the cursor.

```
STAR column updated successfully.
PL/SQL procedure successfully completed.

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
2  CURSOR emp_cur(p_job VARCHAR2) IS
3  SELECT job, sal
4  FROM emp
5  WHERE job = p_job
6  AND sal > 1000
7  FOR UPDATE OF job, sal;
8
9 BEGIN
10 FOR rec IN emp_cur('CLERK')
11 LOOP
12 UPDATE emp
13 SET job = 'SR CLERK',
14     sal = sal + (sal * 0.10)
15 WHERE CURRENT OF emp_cur;
16 END LOOP;
17
18 COMMIT;
19
20 DBMS_OUTPUT.PUT_LINE('Employees promoted successfully.');
```

Employees promoted successfully.
PL/SQL procedure successfully completed.

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10.Display entire record from the Dept table by passing a DEPTNO.Increase the Salary of employees working in deptno 10 by 15% DEPTno 20 by 15% and others 5% Also display the corresponding the employees working in the Dept.Use a parameter Cursor, For Loop Cursor and cursor with Update clause.

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
2  -- Parameter cursor for Department
3  CURSOR dept_cur(p_deptno NUMBER) IS
4  SELECT *
5  FROM dept
6  WHERE deptno = p_deptno;
7
8  -- Parameter cursor for Employees
9  CURSOR emp_cur(p_deptno NUMBER) IS
10 SELECT empno, ename, sal
11 FROM emp
12 WHERE deptno = p_deptno
13 FOR UPDATE OF sal;
14
15 BEGIN
16 -- Display Department details
17 FOR drec IN dept_cur(8DEPTNO)
18 LOOP
19 DBMS_OUTPUT.PUT_LINE('Department No : ' || drec.deptno);
20 DBMS_OUTPUT.PUT_LINE('Department Name : ' || drec.dname);
21 DBMS_OUTPUT.PUT_LINE('Location : ' || drec.loc);
22 DBMS_OUTPUT.PUT_LINE('-----');
23 END LOOP;
24
25 -- Display employees and update salary
26 FOR erow IN emp_cur(8DEPTNO)
27 LOOP
28 DBMS_OUTPUT.PUT_LINE(
29 'Emp No : ' || erow.empno ||
30 ' Name : ' || erow.ename ||
31 ' Salary : ' || erow.sal
32 );
33
34 IF 8DEPTNO = 10 THEN
35 UPDATE emp
36 SET sal = sal * 1.15
37 WHERE CURRENT OF emp_cur;
38
39 ELSIF 8DEPTNO = 20 THEN
40 UPDATE emp
41 SET sal = sal * 1.15
42 WHERE CURRENT OF emp_cur;
43
```

```
44
45 UPDATE emp
46 SET sal = sal * 1.05
47 WHERE CURRENT OF emp_cur;
48
49 ELSIF 8DEPTNO = 30 THEN
50 UPDATE emp
51 SET sal = sal * 1.05
52 WHERE CURRENT OF emp_cur;
53
54 ELSE
55 UPDATE emp
56 SET sal = sal * 1.05
57 WHERE CURRENT OF emp_cur;
58 END IF;
59 END LOOP;
60
61 COMMIT;
62
63 DBMS_OUTPUT.PUT_LINE('Salary updated successfully. ');
64 END;
65 /
Enter value for deptno: 20
old 17: FOR drec IN dept_cur(8DEPTNO)
new 17: FOR drec IN dept_cur(20)
Enter value for deptno: 20
old 10: FOR erow IN emp_cur(8DEPTNO)
new 10: FOR erow IN emp_cur(20)
Enter value for deptno: 10
old 34: IF 8DEPTNO = 10 THEN
new 34: IF 10 = 10 THEN
Enter value for deptno: 20
old 19: ELSIF 8DEPTNO = 20 THEN
new 19: ELSIF 20 = 20 THEN
Department No : 20
Department Name : RESEARCH
Location : DALLAS
-----
Emp No : 7566 Name : JONES Salary : 2975
Emp No : 7788 Name : SCOTT Salary : 3000
Emp No : 7982 Name : FORD Salary : 3000
Emp No : 7789 Name : VAISE Salary : 50000
Salary updated successfully.
PL/SQL procedure successfully completed.
```

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11. Write a program that will prompt the user for a CHOICE. If CHOICE is 1 then display all the employees who earn more than 2000 from EMP table else display all the employees who earn less than 2000. Use REF CURSORS.

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
2   TYPE emp_refcur IS REF CURSOR;
3
4   rc      emp_refcur;
5   v_choice NUMBER;
6
7   v_empno EMP.EMPNO%TYPE;
8   v_ename EMP.ENAME%TYPE;
9   v_sal    EMP.SAL%TYPE;
10
11 BEGIN
12   v_choice := &CHOICE;
13
14   IF v_choice = 1 THEN
15     OPEN rc FOR
16       SELECT empno, ename, sal
17       FROM emp
18       WHERE sal > 2000;
19   ELSE
20     OPEN rc FOR
21       SELECT empno, ename, sal
22       FROM emp
23       WHERE sal < 2000;
24   END IF;
25
26   DBMS_OUTPUT.PUT_LINE('EMPNO  ENAME          SALARY');
27   DBMS_OUTPUT.PUT_LINE('-----');
28
29   LOOP
30     FETCH rc INTO v_empno, v_ename, v_sal;
31     EXIT WHEN rc%NOTFOUND;
32
33     DBMS_OUTPUT.PUT_LINE(
34       v_empno || ' ' || ' ' ||
35       RPAD(v_ename,10) || ' ' ||
36       v_sal
37     );
38   END LOOP;
39
40   CLOSE rc;
41 END;
42 /
```

```
18   WHERE sal > 2000;
19   ELSE
20     OPEN rc FOR
21       SELECT empno, ename, sal
22       FROM emp
23       WHERE sal < 2000;
24   END IF;
25
26   DBMS_OUTPUT.PUT_LINE('EMPNO  ENAME          SALARY');
27   DBMS_OUTPUT.PUT_LINE('-----');
28
29   LOOP
30     FETCH rc INTO v_empno, v_ename, v_sal;
31     EXIT WHEN rc%NOTFOUND;
32
33     DBMS_OUTPUT.PUT_LINE(
34       v_empno || ' ' || ' ' ||
35       RPAD(v_ename,10) || ' ' ||
36       v_sal
37     );
38   END LOOP;
39
40   CLOSE rc;
41 END;
42 /
Enter value for choice: 800
old 12:   v_choice := &CHOICE;
new 12:   v_choice := 800;
EMPNO  ENAME          SALARY
-----
7499    ALLEN           1600
7521    MARD             1250
7654    MARTIN           1250
7814    TURNER           1500

PL/SQL procedure successfully completed.
SQL> |
```

